



NeuroHW IR: A Hardware-Aware Intermediate Representation for Neuromorphic Mapping and Optimization

Thamish Wanduragala | Prof. Milijana Surbatovich | Prof. Manoj Franklin
University of Maryland



01 Motivation

Spiking neural networks (SNNs) communicate through discrete spikes. Event-driven execution can reduce unnecessary computation on neuromorphic hardware.

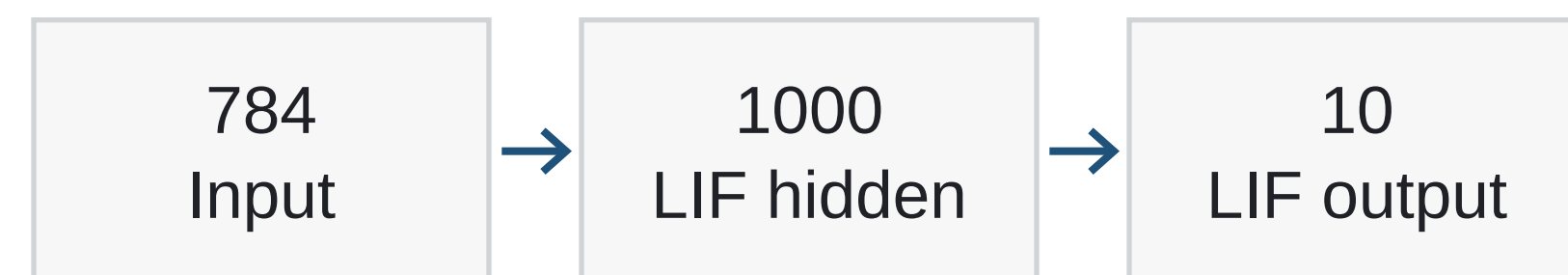
Portable models, hardware mapping, and simulation are often handled separately. This makes it harder to compare and optimize deployment across devices.

The gap
Hardware mapping is outside the model IR.

Goal

Connect portable SNN models to hardware-aware mapping through MLIR. Explicit hardware constraints enable reusable compiler passes and progressive lowering.

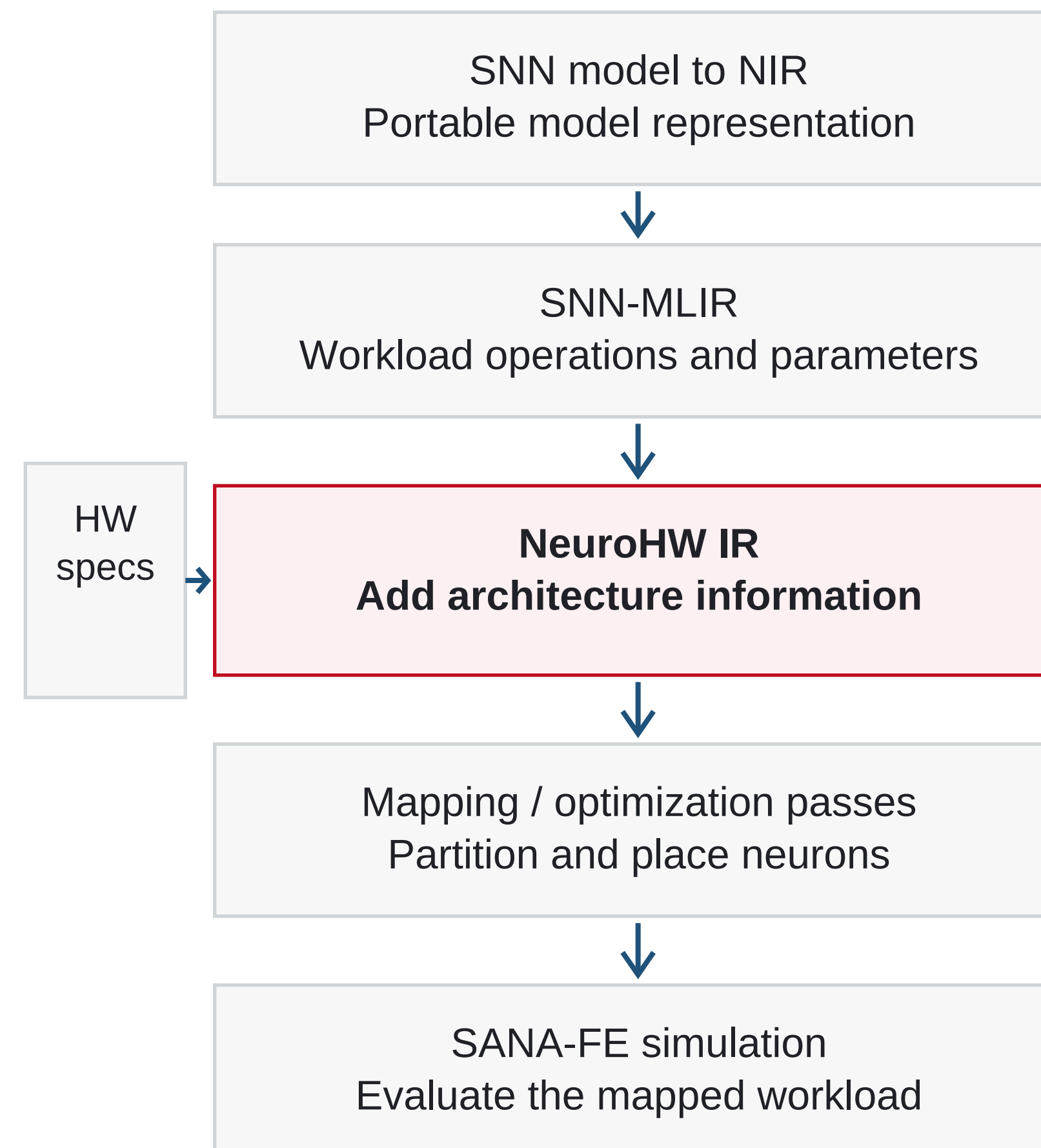
MNIST workload



795,010 weights | 3.03 MB (FP32)
Trained using snnTorch

LIF: leaky integrate-and-fire neurons
28 × 28 grayscale image inputs

02 Proposed workflow



What the IR captures

Initial attributes: cores and neurons per core.
Planned extensions: placement, routing, and mapping constraints.

Evaluation targets
Latency | Energy | Communication

03 Preliminary results

94.33%
MNIST classification accuracy

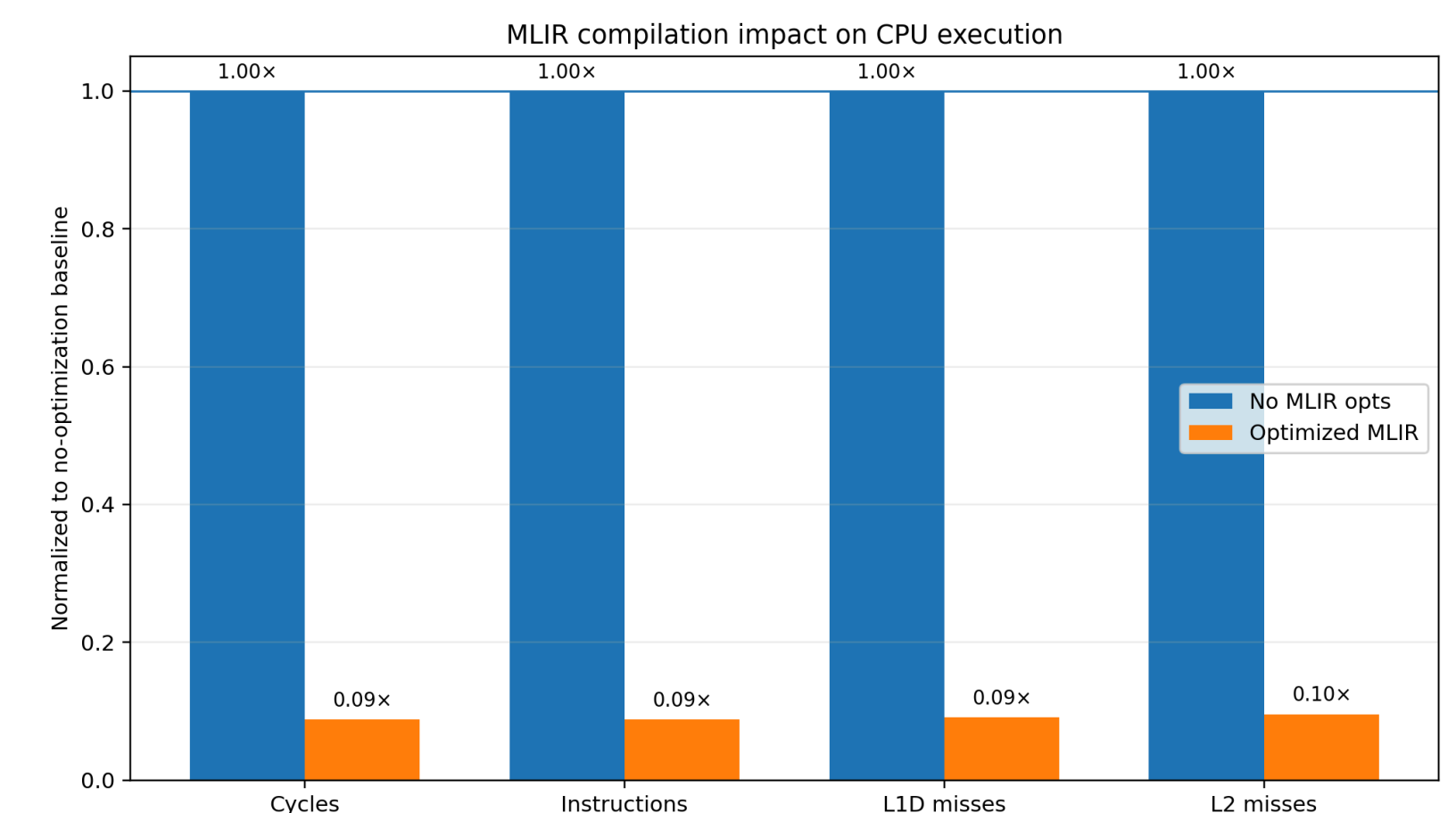
Skipping zero multiplications

Event-driven execution avoids zero work.

23.89× fewer
hidden-layer operations

5.43× fewer
output-layer operations

MLIR optimizations on CPU (gem5)



Scalar replacement | LICM | CSE

Fewer instructions and cache misses with compiler optimizations.

Next steps

Compare mappings in SANA-FE; validate energy, latency, and NoC effects.